

Emily Tan

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EDUCATION

University of California, Riverside

Bachelor of Science in Computer Science (Honors) GPA: 3.96/4.00

Riverside, CA

Expected 2029

EXPERIENCE

ACM Spark Web Developer | Rides | SHPE

Association for Computing Machinery @UCR

Jan. 2026 – Present

Riverside, CA

- Developed scalable UI components using Next.js, TypeScript, and Tailwind CSS for production deployment
- Implemented responsive design patterns to ensure consistent performance across devices and screen sizes
- Collaborated using Git-based workflows to coordinate feature development across a team of engineers

ACM Forge Fullstack Developer | GitHub

Association for Computing Machinery @UCR

Apr. 2026 – Present

Riverside, CA

- Collaborated in a multidisciplinary team developing a smart surveillance system with real-time video detection
- Focused on full-stack development using Next.js, React, and TypeScript to build live monitoring dashboard
- Integrated computer vision streaming pipelines into the frontend using Flask video feeds and MQTT messaging

Computer Science Associate

Formula SAE Highlander Racing @UCR

Oct. 2025 – Present

Riverside, CA

- Programmed STM32 microcontroller to transmit and process CAN bus messages for real-time vehicle systems
- Redesigned control logic using state machines and DMA to improve system speed and reduce CPU load

Embedded Systems & CV Engineer | Scholly Desk Robot

SWE Beehive @UCR

Apr. 2026 – Present

Riverside, CA

- Co-developing an IoT desk robot in a team of 4 using BlazePose for real-time upper-body posture analysis
- Integrating hardware sensors, ambient noise monitors, and a timer manager to automate break reminders
- Programming audio cues, display visuals, and servo gestures across Windows and Raspberry Pi

PROJECTS

UCR Registration Extension | GitHub | Webstore Link

Apr. 2026 – Present

- Built a Chrome extension that aggregates course data from multiple sources into a unified registration interface
- Integrated MyClassGrades, UCR course difficulty data, and RateMyProfessors into a structured data pipeline
- Implemented SPA-aware DOM injection with service workers and Chrome storage APIs for persistent performance
- Deployed and iterated on the extension, reaching **50+ active users**

Pill-E – BioEngineering Makeathon | GitHub

Apr. 2026

- Built the full embedded system for a smart pill dispenser as the sole embedded engineer in a 3-person team
- Developed control logic for precise pill dispensing and real-time device state management
- Implemented device-to-mobile communication layer to interface with a companion app built by teammates
- Debugged and validated hardware performance through iterative testing and calibration of the dispensing system

DevTunes – Google DeepMind Hackathon | GitHub

Mar. 2026

- Worked in a 3-person team to build an AI system that analyzes source code to generate context-aware soundtracks
- Led project documentation, including system design write-ups, workflow diagrams, and technical specifications
- Contributed to designing modular workflows for code parsing, scene detection, and automated prompt generation

ACM DAS Walmart Sales Forecasting | Python, pandas, NumPy, Matplotlib

Jan. 2026 – Mar. 2026

- Built time-series forecasting models including ARIMA, SARIMAX, and XGBoost on retail sales data
- Engineered features capturing seasonality, trends, and holiday effects to improve predictive accuracy
- Evaluated model performance through structured experiments comparing forecasting methodologies
- Analyzed outputs to generate insights on model behavior, accuracy tradeoffs, and data patterns